

DENG - Weekly Schedule HS26

Thursdays, 15:30-17:50

Week / Date / Topic	Theory	Practice
Week 1 - 17.09.2026 Docker & DE Foundations	What is Data Engineering? Data lifecycle; pipeline anatomy; sources -> ingestion -> storage -> transformation -> serving; reproducibility; why containers?	Hands on sessions to design DE lifecycle for different use cases and scenarios.
Week 2 - 24.09.2026 PostgreSQL & Data Ingestion	Source Systems & Storage Design	PostgreSQL in Docker; networking; Python ingestion;
Week 3 - 01.10.2026 Initial Project Pitch: Architecture v0.1	-	Short team presentations: use case, data source(s), data characteristics and challenges and Architecture v0.1.
Week 4 - 08.10.2026 Data Ingestion Patterns & Cloud Infrastructure	Ingestion architecture and design decisions: batch vs. continuous/streaming; full vs. incremental loads; push vs. pull; CDC; ingestion frequency, latency, source-system load, scalability, reliability and failure handling.	GCP + Terraform: provision GCS and BigQuery as ingestion targets; configure providers, variables, resources and state; load data into the cloud landing zone.
Week 5 - 15.10.2026 Workflow Orchestration & Reliable Pipelines	Scheduler vs. orchestrator; workflows, tasks and dependencies; parameters and scheduling; monitoring, bottlenecks and buffers; retries, idempotency, backfills and recovery.	Kestra: build and schedule an ingestion flow; pass inputs/outputs; add retries, variables, backfills, logging and failure handling.
Week 6 - 22.10.2026 Modern Data Platforms & ELT	Why ELT emerged; cloud data warehouses; data lakes and data marts; data lake vs. warehouse; centralized platforms vs. data mesh; integrating ingestion, transformation and quality.	End-to-end data platform lab: ingest NYC Taxi data, transform and validate it, then publish the outputs to BigQuery.
Week 7 - 29.10.2026 Midterm Project Defence	-	Question-only oral defence: 10 minutes per team, including all questions and answers. Both students answer questions about the submitted ingestion/orchestration pipeline, Architecture v0.2, design decisions, reliability and next steps. No prepared presentation, slides or live demonstration. Submit the repository URL and assessed commit hash or release tag through ILIAS by 22.10.2026, 15:30 Europe/Zurich (Week 6).
Week 8 - 05.11.2026 Batch Processing, Spark & Lakehouse	Spark architecture (driver, executors, partitions); DataFrames; lazy evaluation, shuffles, skew, caching; broadcast joins & AQE; data lake -> lakehouse; table formats; Delta Lake (ACID, time travel); partitioning & liquid clustering; Databricks & Unity Catalog.	PySpark basics; Delta tables; Parquet/JSON -> bronze/silver; NYC Taxi analysis; time travel; MERGE & SCD2;
Week 9 - 12.11.2026 Event Streaming & Stream Processing	Kafka (topics, partitions, brokers, producers/consumers, offsets); delivery guarantees; Redpanda; batch vs. stream; stateless vs. stateful; event time, windows, watermarks, late data; checkpointing; backpressure; stream joins; Flink vs. Kafka Streams vs. Spark.	Codespaces (Redpanda, Flink, Dinky); Python producer/consumer; consumer groups; Bluesky Jetstream ingest; windowed aggregations with Flink SQL.
Week 10 - 19.11.2026 Structured Streaming & Incremental Ingestion	Unbounded tables; micro-batch vs. real-time mode; readStream/writeStream; output modes; checkpoints & state; triggers & availableNow; full vs. incremental; high-water marks; Auto Loader; Change Data Feed	Auto Loader with availableNow; HWM ingest; streaming medallion with Kafka (bronze/silver/gold); tumbling, sliding & session windows; CDF as a source;
Week 11 - 26.11.2026 Data Warehousing & BigQuery	OLTP vs. OLAP; why analytics should not run on source systems; warehouse architecture; columnar analytics; external vs. native tables; partitioning, clustering and cost/performance trade-offs.	BigQuery: create external and native NYC Taxi tables; apply partitioning and clustering; compare bytes scanned, query cost and performance.
Week 12 - 03.12.2026 Data modeling	Purpose and levels of data modeling; Inmon, Kimball and Data Vault; facts, dimensions, grain and star schemas; analytics engineering; dbt transformations, dependencies and materializations.	dbt + BigQuery: define sources and staging models; build fact/dimension marts with ref(); add tests and documentation; inspect lineage.

Week / Date / Topic	Theory	Practice
Week 13 - 10.12.2026 Project completion, troubleshooting, and defence preparation		Complete and verify the project pipeline, resolve remaining issues, and prepare to answer questions about the implementation and design decisions. No presentation or slides are required. Final submission deadline: 10.12.2026, 20:00 Europe/Zurich. Submit the repository URL and assessed commit hash or release tag through ILIAS.
Week 14 - 17.12.2026 Final Project Defence		Question-only oral defence: 10 minutes per team, including all questions and answers. Both students answer questions about the submitted end-to-end pipeline, final architecture, implementation, results and trade-offs. No prepared presentation, slides or live demonstration. Submit the repository URL and assessed commit hash or release tag through ILIAS by 10.12.2026, 20:00 Europe/Zurich (Week 13).